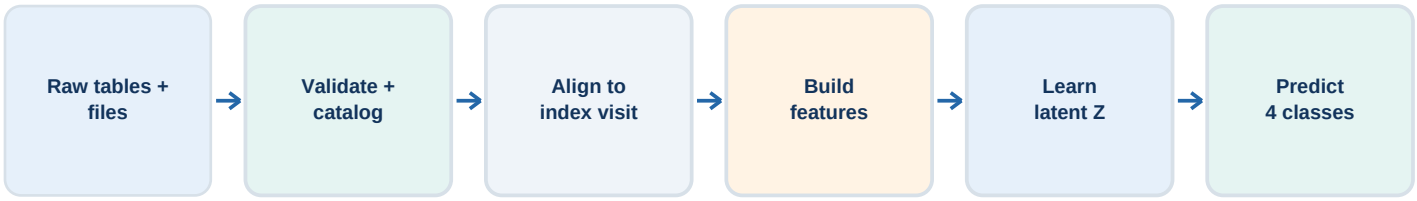
AI-READI

Clinical-first pipeline implementation plan

A complete data path for a present-phenotype prototype, designed so the same processing contracts can later support the full multimodal release.

Phase 1 prediction target: the participant's current four-level study_group. This is a supervised classification head on top of self-supervised clinical representation learning. It is not a future disease-risk, progression, or treatment recommendation model.

The end-to-end plan



BUILD NOW	DESIGN FOR LATER
Use the participant table and OMOP clinical records; collectively, the tables cover all 1,067 people, while individual tables vary in coverage. Pretrain a denoising tabular autoencoder, then train a compact four-class classifier.	Keep raw ingestion manifest-driven and modality-neutral. Add image, ECG, CGM, activity, and environment encoders behind the same participant, time, quality, and availability interfaces.

Decisions fixed in this plan

- One row per participant at the study visit; participant_id is a join key, never a model input.
- Use the provided participant-level train / val / test split, after checking its actual class counts.
- Treat missing modalities and missing clinical answers as missing, with explicit masks and reasons.
- Keep all preprocessing, representation learning, tuning, and calibration inside the training/validation workflow; open test once at the end.

TARGET AND EVIDENCE

1. What this prototype predicts

The response variable is **participants.tsv: study_group**, with four current participant categories. The word “predict” here means infer the assigned group from same-visit data. Because the cohort release is predominantly cross-sectional, it cannot establish who will develop disease later or how an individual will progress.

study_group value	Participants in local participants.tsv	Meaning
healthy	372	No diabetes group in this release
pre_diabetes_lifestyle_controlled	242	Prediabetes or lifestyle-controlled diabetes
oral_medication_and_or_non_insulin_injectable_medication_controlled	323	Type 2 diabetes controlled with oral or non-insulin injectable medication
insulin_dependent	130	Type 2 diabetes controlled with insulin
Total	1,067	Counts recomputed from the local participant table

Current release and local subset

Evidence	What is present	How the plan uses it
Participant / clinical	1,067 participant rows. OMOP person, measurement, observation, condition, procedure, and visit tables collectively cover all participant IDs.	Build the Phase 1 cohort and clinical feature table.
Raw files on disk	Only a small local subset: raw imaging and sensor records represent 13 people; FLIO represents 9. No retinal OCT payload was found locally in the inspected subset.	Exercise parsers and QC only. Do not claim a raw-image/sensor model result from this subset.
Release manifest	The v2.0.0 README describes 165,051 files and about 2.01 TB for the release. A manifest entry is not proof that its file payload is present in this checkout.	Use manifest plus existence, size, and checksum checks before processing each asset.

Count reconciliation required: the local participants.tsv contains 372 healthy and 323 oral/non-insulin participants. The README's split-summary table reports 364 No DM and 331 Oral. Use the row-level participants.tsv for this experiment, regenerate the cross-tab, and record the discrepancy in the data-QC report.

The available split is 747 train, 160 validation, and 160 test participants. Local row counts by group are 372 / 242 / 323 / 130. Because the insulin class is the smallest, use class-aware loss or sampling and report per-class recall; do not report accuracy alone.

DATA INVENTORY

2. Input data and join contract

Source	Format / examples	Phase 1 role	Later multimodal role
participants.tsv	Participant ID, site, study group, age, visit date, suggested split, modality flags	Cohort, target, age, index date, split. Flags are QC metadata only.	Cohort and availability registry.
clinical_data	OMOP CSV: person, measurement, observation, condition, procedure, visit	Primary model inputs from a reviewed allow-list of measurements and survey/SDOH observations.	Clinical modality encoder / tabular view.
retinal_photography	DICOM fundus photographs	No model training from the tiny local payload.	Image encoder; aggregate eyes/scans to participant.
retinal_oct / retinal_octa	DICOM structural OCT and OCT angiography volumes	No local OCT payload; OCTA files are parser/QC examples only.	Volume / slice encoder, then eye and participant aggregation.
retinal_flio	DICOM fluorescence lifetime imaging	Format check only.	Modality-specific image / map encoder.
cardiac_ecg	WFDB header and waveform files	Format check only.	Signal encoder with lead, sampling-rate, and signal-QC metadata.
wearable_blood_glucose	Open mHealth JSON; CGM observations	Format check only.	Time-series encoder or prespecified glycemic summaries.
wearable_activity_monitor	Open mHealth JSON; timestamped activity measures	Format check only.	Time-series encoder or prespecified activity summaries.
environment	ESDS-style CSV with metadata and timestamped sensor channels	Format check only.	Time-series encoder or fixed-window summaries.

Identity and time matching

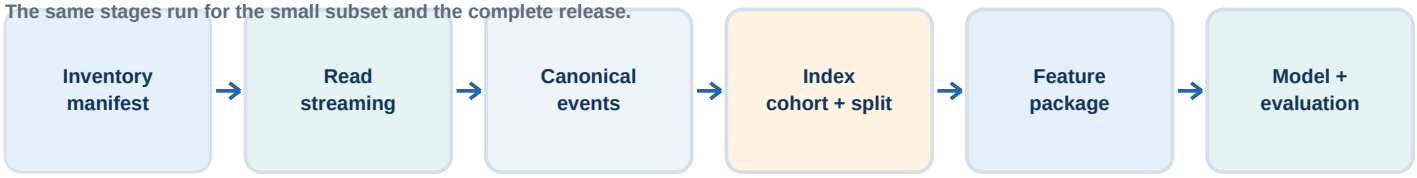
- Normalize participant identifiers as strings without changing their value. Join participants.tsv participant_id to OMOP person_id. OMOP visit_occurrence_id links event rows to visit_occurrence.csv when present.
- Raw files usually sit under a participant directory. Extract and validate the participant ID from that directory and from format-specific metadata (for example Open mHealth patient_id/user_id or environment participant metadata).
- For DICOM and WFDB, catalog participant, modality, device, laterality/lead where available, acquisition time, source path, checksum, and parser/QC status.
- Keep source filenames, source values, original units, and original timestamps in provenance. Never use raw identifiers or timestamps as predictive values.

Input contract: every record must resolve to one participant, one source modality/domain, a source timestamp or explicit unknown-time value, source provenance, and a parse/QC status. Unresolved records go to a quarantine report, not silently into the model table.

RAW DATA TO MODEL-READY COHORT

3. Scalable data flow

The same stages run for the small subset and the complete release.



Stage	Input -> output	Required handling
0. Freeze release	README, metadata, participant table, manifest -> release snapshot	Record release/version, license and use conditions, file count, expected bytes, table schemas, file hashes where supplied. Do not mix dataset releases.
1. Inventory assets	Manifest + filesystem -> asset catalog	Check path, existence, size, checksum, participant, modality, device, format, and duplicates. Report manifest-listed but absent files separately.
2. Parse by type	Raw clinical CSV / DICOM / WFDB / JSON / sensor CSV -> typed records	Stream/chunk large tables. Do not load the 2 TB release into memory. Preserve the original record and parse errors.
3. Normalize	Typed source records -> canonical event records	Retain source and canonical value/unit side by side; convert units only with a recorded rule; attach quality flags and missing reason.
4. Make cohort	Participant registry + events -> one index row per participant	Join by ID; map to the study visit using visit IDs when available, otherwise a fixed, documented date rule. Exclude post-index events.
5. Split first	Cohort rows -> train / validation / test partitions	Freeze participant IDs in each split before any learned transform, feature selection, representation pretraining, augmentation, or tuning.
6. Build features	Allowed index-window events -> model tensors + schema	Create one row per participant; preserve missing masks, modality availability and QC. Fit vocabularies, imputers, scalers and thresholds on train only.
7. Train and report	Train features + labels -> model bundle, predictions, QC and metrics	Select model using validation only. Evaluate test once. Save code/config/data version, feature registry, transforms, seed and model weights.

Intermediate storage

Keep raw assets immutable. Store normalized records and participant-level features in versioned, partitioned columnar files (for example partition by release, domain/modality, and participant or date). Stream large signal/image assets, cache derived features or embeddings, and keep a manifest linking every derived row to its source and transformation version. Make each stage restartable and idempotent: rerunning a parser should update only the same versioned outputs, not duplicate events.

EXACT STARTING INPUTS

4. Phase 1 feature registry

Start with a versioned allow-list rather than every field in every OMOP table. Preserve the original OMOP rows and source columns; create the model view from selected rows only. In measurement.csv, the feature selector is **measurement_source_value** (not a fixed spreadsheet column per lab). The numeric value column is **value_as_number**, with **unit_source_value** retained for unit checks.

Feature block	measurement_source_value examples to review and allow-list
Demographic	participants.tsv: age. Keep clinical_site for audit/subgroup reporting, not as a Phase 1 model input.
Anthropometry	bmi_vsorres, BMI; height_vsorres, Height (cm); weight_vsorres, Weight (kilograms); waist_vsorres, Waist Circumference (cm); hip_vsorres, Hip Circumference (cm); whr_vsorres, Waist to Hip Ratio (WHR).
Vitals	bp1_sysbp_vsorres, Systolic (mmHg); bp1_diabp_vsorres, Diastolic (mmHg); bp2_sysbp_vsorres, Systolic (mmHg); bp2_diabp_vsorres, Diastolic (mmHg); pulse_vsorres and pulse_vsorres_2, Heart Rate (bpm). Preserve repeated readings distinctly until the aggregation rule is set.
Metabolic / renal	Glucose (mg/dL); HbA1c (%); Total Cholesterol (mg/dL); HDL Cholesterol (mg/dL); LDL Cholesterol Calculation (mg/dL); Triglycerides (mg/dL); Creatinine (mg/dL); BUN (mg/dL); Urine Albumin (mg/dL); Urine Creatinine (mg/dL); CRP - HS (mg/L).
Other labs (optional, after coverage/QC audit)	ALT, AST, albumin, total protein, bilirubin, calcium, sodium, potassium, chloride, total CO2, and CBC source values. Each must be explicitly listed with source label, expected unit, validity review, coverage, and missingness before use.

Survey / SDOH observations

observation_source_value allow-list	Encoding and use
years_of_education; cl_maristat	Education and marital status. Confirm codebook; treat codes as categorical or ordered only when the instrument defines an order.
diet1-diet9; dietscore	Diet questionnaire items/score. Keep item responses and score provenance distinct; do not assume score is a validated clinical outcome.
pxfi1-pxfi5	Food-security questions.
pxne1-pxne17	Neighborhood / environment perception questions.
pxed1; pxhi1; pxahc1-pxahc10	Education, health-insurance, and access-to-care question items as available in the local codebook.
pxpa1-pxpa4; sualckncf; susmkncf; suvpkncf1	Physical activity and selected substance-use responses, only if the codebook and leakage review approve them.

For every allowed field, record exact source value, source table, OMOP column used, human-readable label, codebook, data type, unit, expected range, temporal rule, transform, missingness, leakage rationale, and registry version. If an allow-list value is absent or has an unexpected coding, fail the feature build with a QC message; do not quietly substitute a similar-looking field.

RULES THAT PROTECT VALIDITY

5. Alignment, missingness, and leakage

Index and repeated records

- Use study_visit_date as the participant's index date. Prefer visit_occurrence_id to attach OMOP event rows to that encounter. For clinical rows without a usable visit ID, the primary rule is same calendar date as study_visit_date; store the time offset and include no post-index information. Do not widen this window after seeing model results.
- If a measure is repeated at the index visit, retain the repeats or apply a written field-specific aggregation. Examples: keep two blood-pressure readings separately; for repeat lab values use a documented QC-based selection or median. Preserve source rows for audit.
- Do not treat unmatched historical records as contemporaneous predictors. If the visit cannot be aligned confidently, mark the feature missing with reason 'not aligned' and report the count.

Missing values and quality

- Keep separate reasons: not collected, unavailable in local payload, parser failure, invalid/QC-failed, and not aligned to index visit. Keep one mask per feature and one availability/QC record per modality.
- Numeric values: check units and plausible ranges, flag outliers for review, then train-only median imputation plus a missing indicator and robust scaling. Never silently replace an implausible value.
- Categorical values: explicit missing and unknown categories; one-hot or train-vocabulary embeddings. Preserve response codes and codebook labels. A numeric survey code is not necessarily a numeric measurement.
- A false modality flag, missing file, or missing sensor series is not a measured zero. The manifest, participants.tsv flags, and physically present files are separate evidence and must be reconciled.

Leakage exclusions for study_group

Do not feed to model	Why
participant_id/person_id, studyid, UUIDs, filenames, device identifiers, visit IDs, raw timestamps	Identifiers can memorize people or acquisition workflow; timestamps are alignment/provenance only.
study_group and recommended_split	The target and partition are metadata. They are not features.
clinical_site and modality-presence flags	Keep for QC and subgroup reporting. Exclude from the primary model to avoid site/collection patterns becoming shortcuts.
mhterm_dm1, mhterm_dm2, mhterm_predm, mh_dm_age, mh_a1c; diabetes diagnosis rows; diabetes treatment/medication records	Direct diagnosis, group-defining answers, or treatment-status proxies.
cmtrt_a1c, cmtrt_glcs, cmtrt_insln, cmtrt_lfst; paid* and dml* items	Treatment, self-management, or diabetes status can reveal the target directly.
Post-index event records	Information would not be available at the prediction time and could leak outcomes or treatment decisions.

HbA1c and glucose are biological measurements that may be useful predictors, but may also closely reproduce how groups were assigned. Keep a lab-inclusive result and a prespecified sensitivity result without HbA1c/glucose. If the study team confirms either lab defined a label, exclude it from the primary predictor and state that choice.

CLINICAL-ONLY PROTOTYPE

6. Model architecture for the current data

The first model is deliberately small. It uses permitted same-visit clinical features from the cohort and learns a compressed representation before fitting the four-class prediction head. With 1,067 people, the goal is a defensible prototype and reproducible evaluation, not a claim that a large neural network has learned a general clinical foundation model.

A. Model input	B. Self-supervised pretraining	C. Supervised prediction
Numeric features + numeric missing masks Categorical embeddings + categorical missing tokens Feature registry / version	Randomly mask or corrupt observed features Encoder: 256 -> 128 -> 64 Latent representation Z (64-D) Decoder reconstructs the masked inputs	Z -> compact MLP: 32 units -> 4 logits Softmax over study_group Class-weighted cross-entropy

Pretraining objective: reconstruct only values that were observed before corruption.

Training sequence

Step	Operation	Loss / selection
1. Freeze split	Use the 747 train participants to fit every preprocessing statistic and pretrain the encoder. Validation/test participants do not participate in representation learning for this inductive evaluation.	No label used in reconstruction.
2. Pretrain encoder	Corrupt a random subset of present input features. Reconstruct original numeric values with a robust regression loss and categorical values with cross-entropy. Ignore naturally missing cells in the reconstruction loss.	Masked denoising reconstruction.
3. Fit classifier	Attach the four-logit head to Z. Fine-tune on train labels; use class weights or balanced batches because insulin-dependent is the smallest group.	Weighted cross-entropy; train-only features.
4. Select	Try a compact set of hidden sizes, dropout/weight decay, mask rates and early-stopping choices. Select once using validation macro-F1 and check calibration.	Do not select on test results.
5. Compare	Compare against majority-class, regularized multinomial logistic regression and a gradient-boosted tree using the same rows, splits, features and train-only preprocessing.	Model complexity must justify itself against baselines.

At inference, use the saved feature registry and preprocessing objects to produce the feature vector, pass it through encoder and classifier, and return the four probabilities. The reconstruction decoder is a training component; it is not required to predict a class.

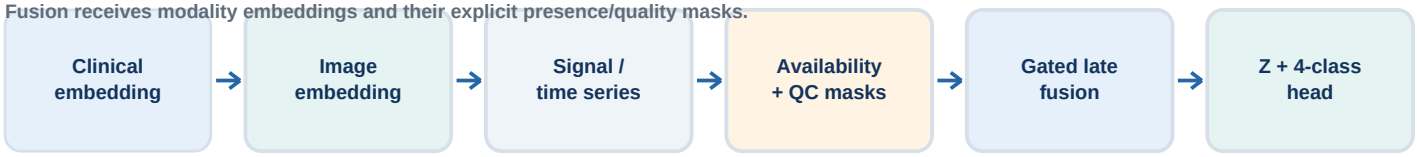
FUTURE FULL-RELEASE MODEL

7. How the same pipeline grows to multimodal

Once the complete payload is locally available and its cross-participant coverage is audited, add one encoder per modality. Each encoder returns a fixed-size embedding plus availability and QC metadata. A missing modality contributes a missing mask, not a synthetic zero image or signal. Late fusion keeps the clinical prototype usable for participants who do not have every modality.

Modality	Preprocess / encoder contract	Participant-level output
Fundus / OCT / OCTA / FLIO	Verify DICOM metadata and pixels; separate modality/device; orient, normalize and QC by device; keep eye and scan grouping. Use a retinal image encoder (compare a suitable pretrained retinal encoder such as RETFound for compatible 2D images). OCT/OCTA volumes need a compatible volume/slice strategy.	Eye embeddings -> participant embedding with mean/attention pooling; retain laterality and scan count.
ECG (WFDB)	Read header and waveform together; validate leads, sample rate, duration and signal quality; resample/filter only with recorded parameters; fixed segment policy.	One or more segment embeddings -> participant ECG embedding and QC summary.
CGM / activity (Open mHealth)	Parse units and timestamps; verify participant key, timezone and sampling gaps; set a predeclared window around index visit; compute safe summaries or encode time series.	Window embedding plus coverage, gap, and valid-duration features.
Environment (ESDS CSV)	Parse metadata before data rows; validate timestamp and channels; stream long files; summarize or encode prespecified windows; retain sensor-QC/gap information.	Window embedding plus coverage and sensor-QC summary.
Clinical tables	Use the Phase 1 event and feature contracts. Optionally retain a clinical tabular encoder output as one modality token.	Clinical embedding and feature-missingness/QC summary.

Fusion receives modality embeddings and their explicit presence/quality masks.



Multimodal self-supervision

- Within modality: masked reconstruction for tabular/signal/time-series data; masked image reconstruction for compatible image inputs; contrastive augmentation only when transformations preserve clinical meaning.
- Across modalities: align embeddings from the same participant and index period, optionally with a contrastive objective. Never align unrelated visits or let a participant's test data influence the learned encoder.
- Use modality dropout during training so the fusion model learns to operate when a modality is absent. Keep a clinical-only path and evaluate each modality ablation.
- Split at participant level before making image crops, eyes, ECG segments, or time windows. All derivatives of one participant stay in one partition.

The raw multimodal branch is a later phase. The 13-person local raw sample is useful for checking parsers, shapes, metadata, and QC output, but is too small to support credible learned imaging or sensor encoders or a multimodal performance claim.

HOW SUCCESS IS REPORTED

8. Evaluation and prediction outputs

Evaluation element	Required decision
Partitions	Use recommended_split at participant level: train 747, validation 160, test 160. Before training, inspect the row-level group-by-split table and document the README count mismatch. Do not move participants based on performance.
Primary metric	Macro-F1 on the locked test set, because each class matters and class frequencies differ.
Also report	Balanced accuracy; per-class precision, recall and F1; confusion matrix with counts and row-normalized rates; one-vs-rest ROC-AUC and PR-AUC where estimable; Brier score or calibration plot; confidence intervals if feasible.
Subgroups / robustness	Report performance by clinical_site and relevant available demographic groups only when sample sizes support it. Include per-class counts and avoid interpreting tiny subgroups. Compare lab-inclusive vs HbA1c/glucose-excluded features; compare model to simple baselines.
Exploratory latent space	UMAP or cluster plots may help inspect Z, but do not constitute predictive validation. Fit visualization parameters on training embeddings where practical; label plots as exploratory and do not claim discovered clinical subtypes from an embedding alone.
Test discipline	Validation may guide tuning and calibration. Lock choices, then evaluate the untouched test split once. Any changes after test inspection require a new held-out evaluation plan.

Per-participant prediction record

Field	Meaning
participant_key	Pseudonymous project key for joining the result inside the permitted analysis environment; never an input feature.
predicted_study_group	One of the four study_group values.
probability_by_class	Four probabilities in a fixed, documented class order; optionally calibrated using validation data only.
latent_embedding	Z vector plus encoder/version identifier; use for exploratory analysis, not as a clinical score.
input_coverage	Which feature blocks/modalities were available, missing reasons, index date alignment status, and QC summary.
provenance	Dataset release, participant split, feature-registry version, preprocessing/model version, and inference timestamp.

The prototype output is a research prediction of the dataset's current study_group. It is not an ocular outcome, systemic complication risk, future progression estimate, diagnosis, or clinical recommendation. Those outcomes require their own labels, index dates, and time-separated evaluation.

WORK PLAN

9. Implementation order and completion gates

Phase	Work product	Gate before advancing
A. Data audit	A versioned data inventory: participant counts, labels/splits, table schemas/row counts, file manifest vs present payloads, join success, modality coverage, and class-count reconciliation.	All 1,067 participant IDs accounted for; unmatched rows and missing files quantified; README-vs-participant count discrepancy logged.
B. Canonical processing	Restartable parsers; event table; asset catalog; unit mapping; codebook maps; QC and quarantine reports; tests of each parser on local examples.	Source records link back to inputs; parser errors and unit conversions are visible; rerunning does not duplicate rows.
C. Cohort/features	Frozen index-visit rule and split; allow-listed one-row-per-person feature package; schema and feature registry; missingness and leakage report.	Exactly one row per eligible participant; no post-index events, IDs, target, split, or direct label proxies in model matrix.
D. Current model	Train-only imputers/scalers/vocab; denoising autoencoder; four-class head; baseline models; saved model bundle.	Training completes reproducibly; validation selects model; test is not used for tuning; class order and probability outputs verified.
E. Evaluation	Locked test metrics, confusion matrix, calibration, subgroup counts, HbA1c/glucose ablation, and model card.	Results include all requested metrics and limitations; no future-risk or causal claims.
F. Multimodal scale-up	Per-modality parser/encoder, embedding cache, availability-aware fusion, modality ablations, and cohort coverage report.	Only start after full payload access and enough participants with usable modality data; split remains participant-level.

Artifacts to deliver with the finished project

- Data inventory and source-to-feature dictionary.
- Versioned participant feature table plus data-QC, join, missingness, and leakage reports.
- Training configuration, preprocessing objects, encoder/classifier weights, and reproducible predictions.
- Evaluation report with baselines, locked test metrics, subgroup counts, lab ablation, and exploratory latent-space plots.
- Model card stating target definition, intended research use, data release, exclusion rules, evaluation, known limitations, and that the output is not for clinical decisions.

Limitations to carry into the final report

The local clinical subset is a usable starting point for the current four-class prototype, but small sample size and label imbalance make overfitting possible. The local raw imaging/sensor payload is far too small for multimodal training. The current target reflects present recruitment/phenotype categories, not incident disease. The release README and participant table disagree on two group counts, so row-level counts are authoritative for this run until the discrepancy is resolved.

USE ALONGSIDE THE PLAN

10. References and practical resources

Resource	Why it helps
AI-READI dataset v2.0.0 README and local metadata	Primary source for release scope, directory layout, data standards, access/license notes, suggested participant split, and the row-level files inspected for this plan. Local source: Part_1/README.md, participants.tsv, participants.json, file-manifest.tsv, clinical_data/.
AI-READI documentation: OMOP Clinical Data Structure	Explains how participant IDs link across person and event tables, why OMOP data are long-form, and how users can create an analysis-specific feature table. docs.aireadi.org - OMOP structure
Owsley et al. Cross-sectional design and protocol for AI-READI. BMJ Open, 2025;15:e097449.	Study design, data-collection domains, and temporal limitations for the cohort. doi:10.1136/bmjopen-2024-097449
Zhou et al. A foundation model for generalizable disease detection from retinal images. Nature 622, 156-163 (2023).	A reference for future masked-image pretraining and retinal encoder adaptation. It is a candidate resource for a later retinal-image branch, not a requirement for the clinical-only Phase 1 model. doi:10.1038/s41586-023-06555-x
OHDSI Book of OHDSI	Background on OMOP conventions and analytic use of standardized clinical tables. ohdsi.github.io/TheBookOfOhdsi

Terms used in this plan

Term	Meaning in this project
Index visit	The study visit that defines when inputs are considered available for a participant's prediction.
Feature registry	Versioned catalog of every allowed input, its source, unit/codebook, alignment and transformation rules.
Latent representation Z	A compact vector learned by the encoder from the allowed input features.
Leakage	Information in the inputs or training process that reveals the answer, uses future information, or crosses the participant split.
Availability-aware fusion	Combining modality embeddings while telling the model which modalities are actually present and pass quality checks.

Scope of this document: a complete implementation plan only. No data-processing pipeline or model code has been written or changed.